



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI KANNIGA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

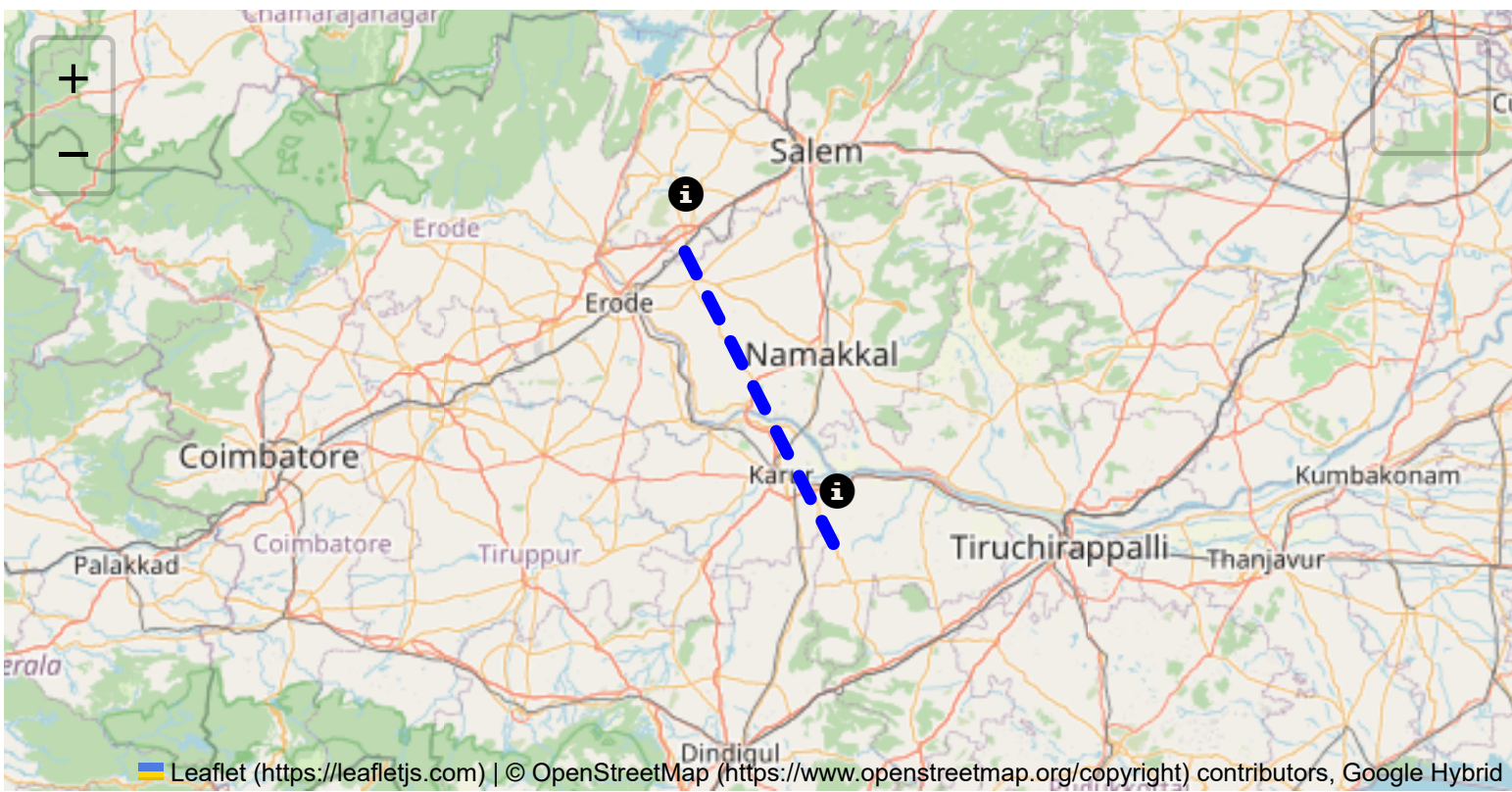
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

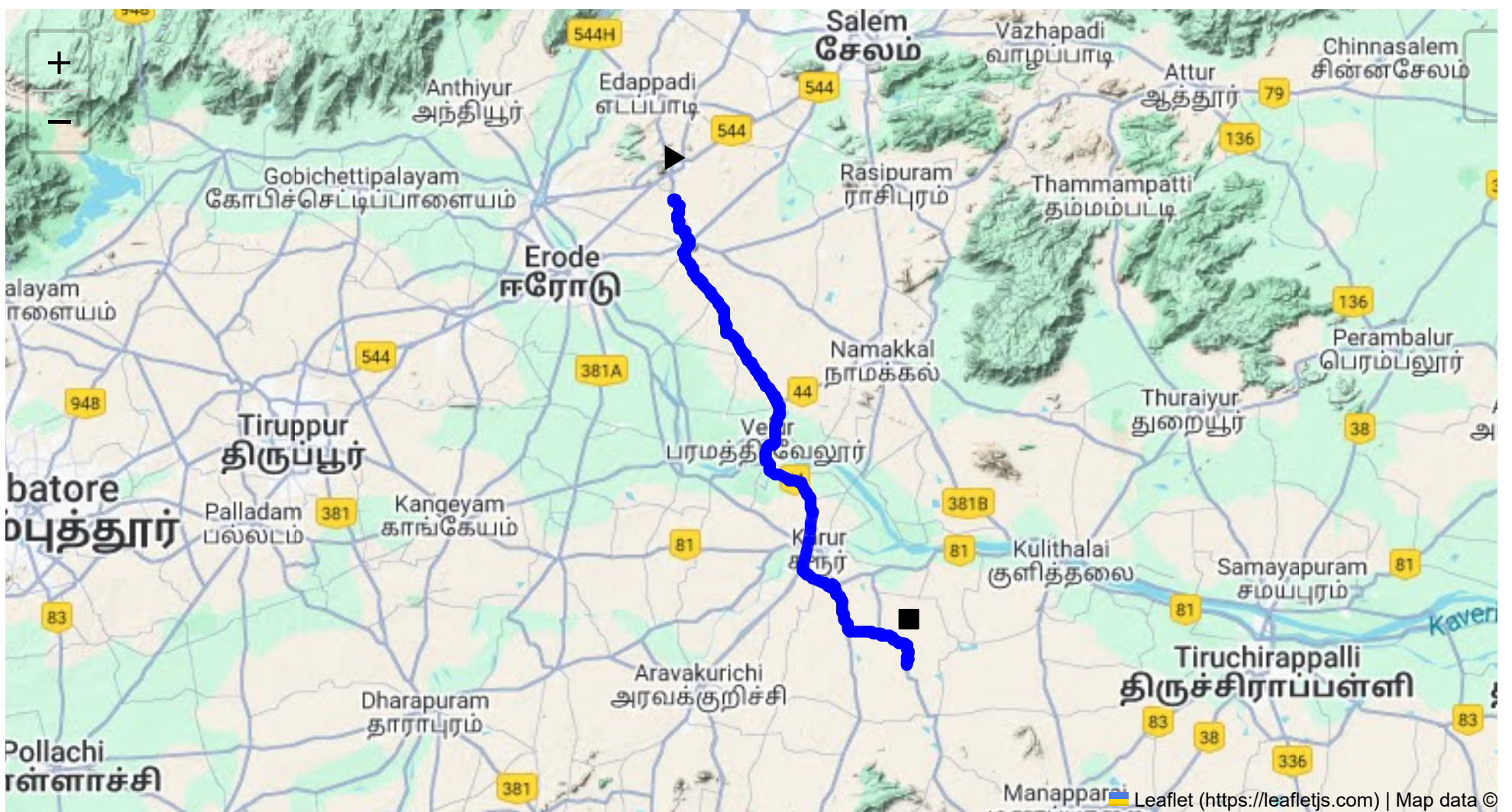
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 94.86 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (10.802598, 78.201081)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri, Tamil Nadu to Manjanaickenpatti, Tamil Nadu

1. Overview of the Route Map

- The route from Sangagiri to Manjanaickenpatti covers approximately 94.86 kilometers, predominantly via regional highways and state routes. The pathway navigates through both rural and semi-urban areas, offering limited bypass routes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- The region experiences a tropical climate, with hot summers and seasonal heavy monsoons, particularly from June to November. Drivers should be cautious of flooding and waterlogging in low-lying areas during the monsoon. Visibility might be reduced during heavy rainfall, and roads can become slippery.

3. Traffic Patterns and Peak Hours

- Traffic is relatively heavier during morning (7:00 AM - 9:00 AM) and evening peak hours (5:00 PM - 7:00 PM) due to local commuter traffic. Market towns and junctions along the route typically experience congestion, especially near Sangagiri and Salem.

4. Assessment of Road Quality and Infrastructure

- The state highways are generally in fair condition but can vary widely, with some stretches having potholes and narrow shoulders. Pay close attention to poorly lit sections and sudden curves, especially during night driving.

5. Suggestions for Alternative Routes for Emergencies

- In case of emergencies, a possible alternative is taking the NH44 briefly, which is better maintained and more direct, though it may add to travel time and require detours to connect back to state highways.

6. Local Regulations Affecting Hazardous Material Transport

- Tamil Nadu has stringent regulations for the transport of hazardous materials, including mandatory permits and adherence to specified routes. Nighttime travel might be restricted in some areas for trucks carrying hazardous materials.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- There have been several incidents, primarily due to speeding and overloading. There is an increased risk of accidents at intersections and in poorly lit areas.

8. Environmental Considerations and Sensitive Areas

- The region encompasses agricultural areas that could be affected by hazardous spills. Care should be taken to avoid spills, especially on fertile land and near water sources.

9. Communication Coverage and Potential Dead Zones

- Mobile coverage is generally reliable along the main route, but expect possible weak signals in rural stretches and near forested areas. Satellite phones or alternate communication means are advisable for uninterrupted contact.

10. Estimated Emergency Response Times for Different Route Segments

- Closer to urban centers like Salem, emergency response times can range between 15-30 minutes. In more remote sections, response times may extend to over an hour due to limited infrastructure.

11. Overall Summary of Risk Assessment

- The route involves moderate risk for heavy vehicles carrying hazardous materials due to variable road conditions, traffic congestion, and weather-related challenges. Adherence to local

regulations and emergency preparedness are crucial for safe transit. Travelers should remain vigilant for hazards, maintain safe speeds, and ensure communication lines are operational throughout the journey. Emergency and alternative routes should be planned in advance, considering the current weather forecasts and road conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	U-Turn	High	10.9110477, 78.09713769999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	Medium	11.10102, 78.00572	30 KM/Hr
12	Turn	Medium	10.90767, 78.09893	30 KM/Hr
13	Turn	High	10.90793, 78.09916	15 KM/Hr
14	Blind Spot	Blind Spot	10.91105, 78.09714	10 KM/Hr
15	Turn	High	10.91103, 78.09733	15 KM/Hr
16	Turn	High	10.84613, 78.12000	15 KM/Hr
17	Turn	Medium	10.84321, 78.16436	30 KM/Hr
18	Turn	High	10.84057, 78.16518	15 KM/Hr
19	Turn	High	10.83953, 78.18108	15 KM/Hr
20	Turn	Medium	10.83531, 78.18158	30 KM/Hr
21	Turn	Medium	10.82892, 78.20166	30 KM/Hr
22	Turn	Medium	10.82865, 78.20188	30 KM/Hr
23	Blind Spot	Blind Spot	10.80223, 78.20131	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 10.9110477, 78.09713769999999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10102, 78.00572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.90767, 78.09893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.90793, 78.09916



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.91105, 78.09714



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.91103, 78.09733



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.84613, 78.12000



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.84321, 78.16436



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.84057, 78.16518



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.83953, 78.18108



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.83531, 78.18158

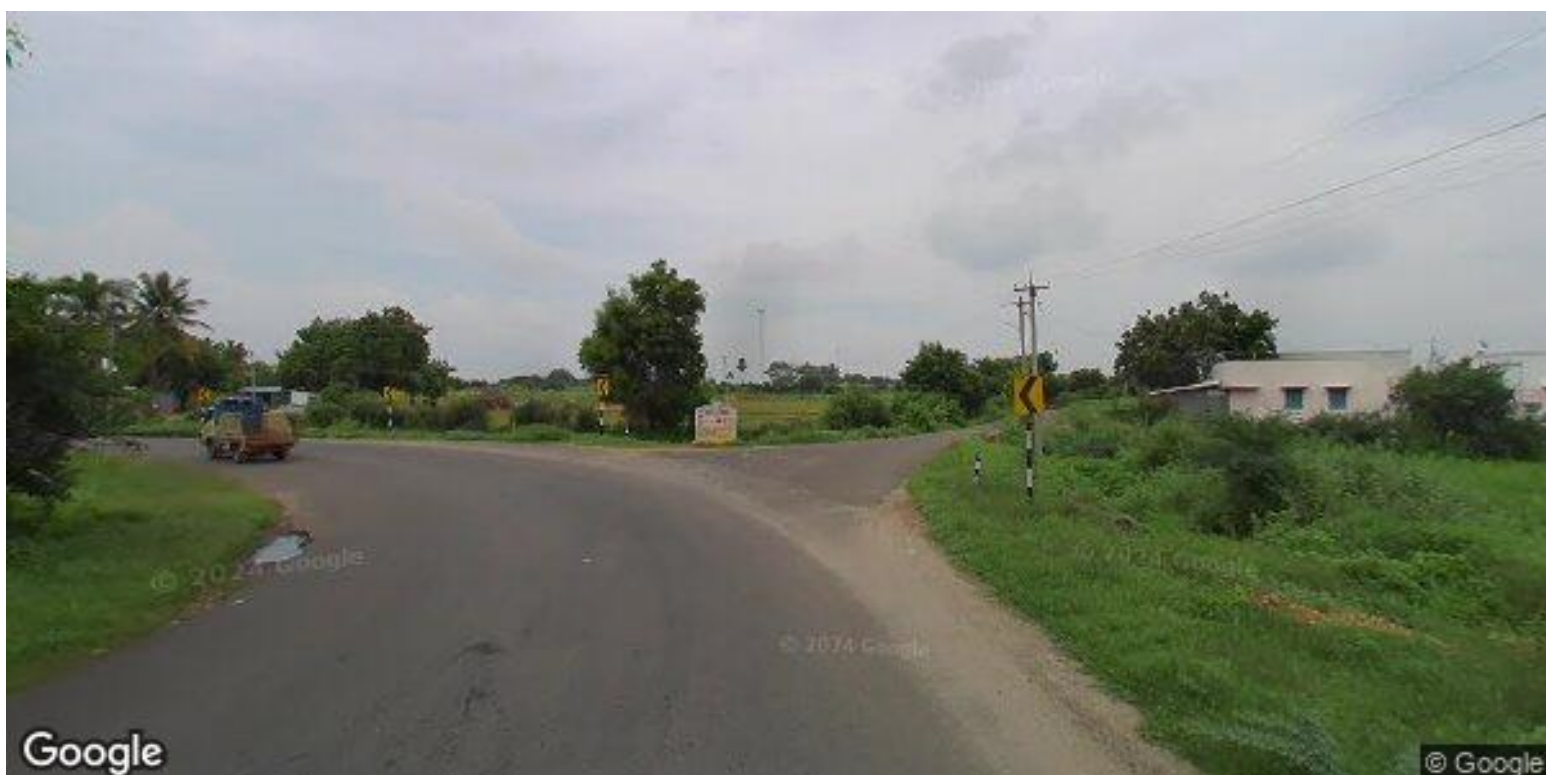


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.82892, 78.20166



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.82865, 78.20188



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.80223, 78.20131